Simultaneous measurement of the ν_μ charged-current inclusive cross section on hydrocarbon in five observables from MINERvA open data

Joseph Bailey[1,*] and Benjamin Nachman[2,3,†]

[1] *Department of Physics, Stanford University, Stanford, California 94305, USA*
[2] *Department of Particle Physics and Astrophysics, Stanford University, Stanford, California 94305, USA*
[3] *SLAC National Accelerator Laboratory, Menlo Park, California 94025, USA*

(Dated: November 4, 2026)

We measure the inclusive muon-neutrino charged-current cross section simultaneously in five observables: muon transverse and longitudinal momenta, available hadronic energy, three-momentum transfer and hadronic invariant mass. MINERvA medium-energy open data are unfolded with OmniFold without binning and reported in sixteen cells. The median relative uncertainty per cell is 7.86%, including a run-period contribution estimated after differences between periods were observed. As a descriptive comparison, three external generator predictions give χ^2 values of 107–229 for 16 degrees of freedom; GiBUU gives 48 on the eight cells covered by its sample. The unfolding reproduces the published muon-kinematic cross-section integral within about 1%, with a correlated shape difference. Fake-data studies find residual bias exceeding uncertainty-based thresholds fixed in advance at a GiBUU-like truth and a synthetic momentum-transfer tilt. The coverage of the reported uncertainty intervals has not been validated.

I. INTRODUCTION

Multidifferential neutrino cross sections test interaction models across muon and hadronic observables. Their interpretation depends on correcting detector response and on assessing the bias and uncertainty of the unfolding procedure. Iterative Bayesian unfolding on a binned migration matrix [1] is widely used, but its response tensor grows rapidly as observables are added. OmniFold learns event weights in an unbinned feature space through classifier-based density-ratio estimation [2, 3]. This makes simultaneous higher-dimensional unfolding possible without specifying a separate binned response tensor for every projection. Unbinned unfolding has been applied to collider data, for example to eight observables in deep-inelastic scattering at H1 [4] and to twenty-four Z+jets observables at ATLAS [5] (see Ref. [6] for a survey). In neutrino physics, it has been demonstrated on a public simulated T2K near-detector sample [7]. Multidifferential neutrino measurements on data reach three dimensions [8–11].

We apply the method to MINERvA medium-energy ν_μ data from the open-data release [12]. The result is a simultaneous charged-current inclusive cross section in five observables: the transverse and longitudinal muon momenta p_T and $p_\parallel$, the available hadronic energy E_{avail}, the three-momentum transfer q_3, and the hadronic invariant mass W. It is reported in sixteen cells with its covariance (Sec. V) and compared with five generator predictions (Sec. V B). We first reproduce the published double-differential cross section $d^2\sigma/(dp_T\, dp_\parallel)$ [13] as an end-to-end test of the unfolding. The three- and four-observable central values are given in the companion note [14]. Reconstructed W is fixed by the reconstructed $(p_T, p_\parallel, q_3)$, so the five observables are simultaneous but not independent.

The validation studies assess the result through the two-dimensional reproduction, recovery of generator-sized truth changes, a coverage test, and a fake-data study of estimator bias. They examine whether the unfolding gives reliable multidimensional cross sections when the simulation mismodels the truth by generator-sized amounts.

Interval coverage has not been validated. The bias tests, the fake-data study, and the dependence of the generator χ^2 on the flux correlations are reported for the sixteen-cell result in Secs. IV G and V B. The findings are those of one OmniFold implementation on one data set, for the observables, configurations, and deformations tested, rather than of multidimensional OmniFold in general.

Section II describes the data, the simulation and the observables, and Sec. III the unfolding and the reporting cells. Section IV gives the uncertainties and the validation studies, from the reproduction of the published two-dimensional measurement to the fake-data tests of the sixteen-cell result, and Sec. V gives the result and its comparison with generator predictions. Section VI discusses what limits the estimator, and Sec. VI A summarizes the principal limitations. The companion analysis note [14], cited below by section, gives the full validation record, the construction of the uncertainties, and the tests with their assumptions.

II. DATA, SIMULATION, AND OBSERVABLES

The data come from the twelve MINERvA medium-energy, forward-horn-current run periods (playlists 1A–1G and 1L–1P), corresponding to 1.057×10^{21} protons on target, with an average neutrino energy near 6 GeV [13].

* josephrb@stanford.edu
† nachman@stanford.edu

The input is an earlier production of the MINERvA open-data release (see Data availability), in the form of pre-selected analysis ntuples (AnaTuples) [12]. The detector, including the downstream MINOS near detector that serves as a muon spectrometer, is described in Ref. [15]. The signal is a ν_μ charged-current interaction, with any final state, whose true vertex lies in the tracker fiducial volume ($5980 < z < 8422\,\mathrm{mm}$, inside a hexagon of apothem 850 mm). The signal muon has $\theta_\mu < 20°$, $1.5 < p_\parallel < 60\,\mathrm{GeV/c}$, and $p_T \leq 4.5\,\mathrm{GeV/c}$. Selected simulated events outside this definition, including ν_μ charged-current events whose muon lies outside these limits, are background. The selection requires a reconstructed vertex in the same volume and a reconstructed muon with $\theta_\mu < 20°$ matched to a MINOS track of negative charge, together with a dead-time requirement, as in Ref. [13]. The flux treatment follows Ref. [16]. For the results in more than two observables, the integrated flux is $\Phi = 6.243 \times 10^{-8}\,\mathrm{cm^{-2}}$ per proton on target, from 0 to 100 GeV, after the neutrino–electron-scattering constraint and averaged over the twelve playlists by exposure (Ref. [14], Sec. 3.4). The analysis uses 4.12 million selected data events. The simulation contains 32.8 million unweighted true signal events and 20.5 million selected simulated events, and it is scaled to the data exposure. The simulated events are generated with GENIE 2.12.6 [17] and reweighted with the MINERvA Tune v1, which applies a non-resonant pion reduction, a random-phase-approximation correction, and an empirical low-recoil two-particle–two-hole (2p2h) enhancement [18]. Event selection is held fixed as the classifier inputs and unfolded observables grow from $(p_T, p_\parallel)$ to $(p_T, p_\parallel, E_{\mathrm{avail}}, q_3, W)$. Adding features does not add selected events or relax the signal definition.

The unfolded observables are defined at truth level, with muon momentum components $p_T = p_\mu \sin\theta_\mu$ and $p_\parallel = p_\mu \cos\theta_\mu$. The magnitude of the three-momentum transfer $q_3 = |\vec{q}|$ comes from the generator's four-vectors, and the hadronic invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ uses the struck-nucleon mass M. We define the available energy as

$$E_{\mathrm{avail}} = \sum_p (E_p - m_p) + \sum_{\pi^\pm} (E_\pi - 135\,\mathrm{MeV}) + \sum_{\pi^0} E_\pi + \sum_\gamma E_\gamma, \tag{1}$$

which subtracts a defined constant of 135 MeV, rather than the charged-pion mass, in the charged-pion term. The constant defines the reported observable and is used consistently by the analysis and every prediction. The sums run over the generator's final-state particles after final-state interactions, where neutral pions appear undecayed. All other final-state particles, including neutrons, contribute nothing. No single published definition of E_{avail} exists. Eq. (1) keeps the species of the closed list of Ref. [18] except electrons and positrons, omits species such as kaons that the open list of Ref. [19] includes, and has a charged-pion term 4.57 MeV per pion above the kinetic energy used in Refs. [18, 20] (Ref. [14], Sec. 7.1). To first order in the unfolding, without retraining, replacing the constant alone by the charged-pion mass moves the sixteen-cell result by at most 0.19σ in any cell and by 0.42σ as one coherent shift, the norm of the shift under the reported covariance (0.21σ and 0.42σ without the run-period block). Reconstructed E_{avail} is 1.17 times the calorimetric recoil energy in the tracker and electromagnetic calorimeter, with muon deposits removed. Reconstructed q_3 and W instead combine the muon kinematics with a separate calorimetric estimate q_0 of the energy transfer.

Reconstructed W is an exact function of the reconstructed $(p_T, p_\parallel, q_3)$, apart from a clamp at $W^2 = 0$, because $W^2_{\mathrm{reco}} = (\bar{M} + q_0)^2 - q^2_{3,\mathrm{reco}}$, with $\bar{M}$ an average nucleon mass, and $q_{3,\mathrm{reco}}$ increases with q_0 at fixed muon kinematics. The five reconstructed inputs therefore share information and do not provide five independent constraints. Only reconstructed W is fixed by the others; reconstructed E_{avail} comes from its own calorimetric estimator and is not a function of them. At truth level, θ_μ, and with it p_T and $p_\parallel$, is measured from a fixed beam axis rather than from the true neutrino direction, so truth W is fixed by $(p_T, p_\parallel, q_3)$ only up to that event-by-event angle and the choice of M. The definitions are detailed in Ref. [14], Secs. 7.1 and 8.2.

The external predictions for CH are the untuned GENIE 2.12.10 without ("GENIE CV") and with ("GENIE + MEC") the Valencia 2p2h component [21], which models meson-exchange currents (MEC). NuWro 21.09 [22] and GiBUU 2019 [23] are generated on carbon; in the sixteen-cell comparison of Sec. V B they are corrected to CH by adding each generator's own free-proton prediction with weight 1/13 per nucleon. All predictions are per nucleon of their target: CH, or carbon for NuWro and GiBUU outside the sixteen-cell comparison. The GENIE and NuWro predictions are reweighted exactly, event by event, to the analysis flux. The GiBUU sample stops at $E_\nu \approx 20\,\mathrm{GeV}$. In the sixteen-cell comparison the predictions use the analysis definitions of W and of the muon angle: W is computed with the struck nucleon's mass, or an average nucleon mass where the generator records no single struck nucleon, and the muon angle is measured from the beam axis, with each event's beam-direction offset sampled from the analysis simulation's distribution. This changes no cell of any prediction by more than 0.07 of the cell's uncertainty (at most 0.063). Elsewhere the predictions compute W with the neutron mass for every event and measure p_T about their own neutrino axis. Hydrogen has no quasielastic, 2p2h, or coherent ν_μ charged-current channel, so normalizing per nucleon does not remove the difference between the targets. In GENIE's own CH sample, the carbon-only cross section per nucleon exceeds that of CH by 8.3 % for $W < 1.1\,\mathrm{GeV}$. The comparisons of NuWro and GiBUU on carbon contain a target difference of this kind. The comparisons of the predictions with the analysis definitions are in Ref. [14],

Secs. 8.1 and 8.3.

The fine grid is the analysis binning: the published 14×16 muon-momentum bins, crossed with 7 E_{avail}, 7 q_3, and 6 W bins. A coarse 3^5 grid in $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ has its edges at the fine edges nearest the simulated tertiles. A coarse cell is kept if its expected reconstructed signal and its truth weight at the data exposure are both at least 1000 events. We call this requirement the support rule. The 109 coarse cells that meet it are the joint cells, on which the note's joint tests are evaluated. The others, which hold 0.47 % of the expected signal, stay in the total only. The construction of the coarse grid is given in Ref. [14], Sec. 11.1.

III. CROSS-SECTION EXTRACTION

Each OmniFold iteration has two classification steps. A reconstruction-level classifier distinguishes data from simulated events and defines a likelihood-ratio weight. The event pairing transfers that weight to truth, including unmatched events, and a truth-level classifier propagates a smooth reweighting to the next iteration. Cross sections are formed by histogramming the reweighted truth events in the reporting binning and dividing by integrated flux, target count, and bin width. Events that fail reconstruction enter as unmatched truth events, so no separate efficiency correction is applied.

Backgrounds enter the reconstruction-level step through the density the classifier learns to match. In the purity-based subtraction, every data event in reconstructed bin b carries the weight $w_{\text{pur}}(b) = \max(0,\, 1 - B_b/D_b)$, where D_b and B_b are the data and predicted background counts in that bin, so the classifier's target density is $D(x)\, w_{\text{pur}}(b)$. The unbinned negative-weight treatment instead adds simulated background events with negative weight to the data, so that the target is $D(x) - B(x)$ pointwise. A classifier refinement [24], clipped at zero, converts the signed sample to non-negative weights before the first iteration. The two targets share bin integrals wherever $B_b < D_b$ but differ inside a bin wherever the background fraction varies within it (Ref. [14], App. C).

The results use two estimators. Both use five OmniFold iterations, gradient-boosted decision trees [25, 26] of nominal capacity (100 trees, 8 leaves) in both OmniFold steps, and negative-weight background subtraction with a refinement classifier at 400 trees and 31 leaves. The two-observable estimator, LightGBM [26] at a fixed seed, gives the two-dimensional central value and its covariance (Sec. IV B). This reproducible central value is that of its own systematic-universe ensemble, so its covariance is built with the same estimator. The three- and four-observable central values and the sixteen-cell five-observable result (Sec. V) use the joint-test estimator, a deterministic LightGBM with the same background treatment and refinement capacity, named for the note's calibrated joint tests of generator predictions, which use it (Ref. [14], Sec. 11). Its seeds are fixed, so its training is reproducible, and it also gives the held-out shape tests of Fig. 2b. This estimator removes most of the bias of purity-based subtraction at nominal truth in the highest-W cells (Ref. [14], Sec. 9.2). Results from the two estimators are not combined. The other configurations used below, and their results, are described in Ref. [14], Sec. 3.2.

The iteration count and classifier capacity regularize the unfolding. No learner setting, whether capacity, learning rate, refinement capacity or iteration count, was tuned on data. The nominal settings are the scikit-learn defaults, with LightGBM matched to their capacity. The refinement capacity, which the two estimators share, was chosen in simulation closure studies, and so were the joint-test estimator's five iterations (below). The two-observable estimator uses five iterations following common practice; this choice was checked on data, against ten iterations with purity-based subtraction and against the published result, without a search over the iteration count (Ref. [14], Sec. 3.2). Which learner implementation gives the two-dimensional central value, at these settings, was chosen after the implementations had been compared with the published result (Sec. IV B). Each classifier is trained on the full sample whose weights it then sets, so no held-out or cross-fitted split is used; it is trained without early stopping, and the two classes keep their physical weights (Ref. [14], Sec. 3.1).

The joint-test estimator keeps five iterations because 10, 15, or 20 iterations did not reduce its model dependence, measured as the largest residual proxy over the truth changes used in its development. The comparison and its values are given in Ref. [14], Sec. 9.4.

A. Reporting cells

A five-observable cross section with total uncertainties is reported as the joint-test estimator's data result on 16 cells, R00–R15. Other reporting definitions of a five-observable measurement, with their precision, are studied in Ref. [14], Sec. 9.4. They come from the 32 boxes formed by one interior edge per axis, at p_T 0.7, $p_\parallel$ 4.5 and q_3 2 GeV/c, and at E_{avail} 0.8 and W 1.8 GeV. In forming the cells, each supported box with purity below 0.1 is merged into a neighbour, 4 boxes remain unreported, and every reported cell passes the support rule. The result's covariance is given in Sec. IV A.

IV. UNCERTAINTIES AND VALIDATION

A closure test measures how well a specified deformation of the known simulated truth is recovered, with the prior, the initial truth distribution, held fixed. Coverage, how often intervals contain their target over repeated experiments, tests the estimator and the interval construction together. The closure, deformation, and coverage studies reweight the true signal of the simulated events used as pseudo-data, which changes their reconstructed

and true distributions together and keeps the detector response at fixed truth. Throughout, the prior used to unfold them stays at Tune v1. Every study uses the analysis simulation's detector response.

A. The reported covariance

The reported covariance of the sixteen-cell result is $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$. The term C_{total} sums the statistical, flux, interaction, detector, lateral and normalization uncertainties of the joint-test estimator and C_{model}; the lateral uncertainties shift the reconstructed muon kinematics and beam angle rather than reweighting events (Appendix A). The model term C_{model} is built from the change in the unfolded data when the prior is reweighted to GENIE + MEC, NuWro and GiBUU in turn. Its contributions come from those generators' samples in their own definitions, with NuWro and GiBUU on carbon and GENIE CV as the common reference, not from the predictions compared in Sec. V B. The comparisons there with GENIE + MEC, NuWro and GiBUU therefore use covariance contributions derived from the same generators. It is a measure of unscaled generator-shape dependence, not a calibrated bound (Ref. [14], Sec. 10.3). The response term C_{resp} is the response of the result to a one-standard-deviation scale of the reconstructed recoil energy in the simulation. The scale is applied to proton, meson, electromagnetic and other deposits. C_{resp} does not cover neutron or muon deposits, nor species, energy-dependence or resolution effects beyond a scale. It enters only this result, its fake-data tests and its comparison in Sec. V B. The unfolded results of the three run-period groups, each relative to its own simulation, differ by more than their bootstrap scatter. The offsets of the twelve playlists differ between the three period groups by more than expected from their playlist-to-playlist scatter (Ref. [14], Sec. 10.4). The cause is not known, and these data cannot establish which period is correct. The run-period block $C_{\text{PL,5D}}$ includes this inconsistency in the reported covariance. It is built from the double ratios among the groups, each group's unfolded result over its own simulated truth, with no group chosen as the reference, so it contains the differences by construction. It was added after these differences were seen, following an audit whose stopping condition was fixed before any of its numbers were read. It has rank 2 and is not corrected for the bootstrap noise it contains, and with three groups its size is poorly determined: the 68 % interval on the ratio of its true to its estimated size is 0.73–2.56. The median per-cell relative uncertainty is 7.86%, 7.29% without $C_{\text{PL,5D}}$, and 6.74% without $C_{\text{PL,5D}}$ or C_{resp}. The block raises the median by about 7.8 %, and the uncertainty of the most affected cell, R04, by 14.1 %. The fake-data tests (Sec. IV G) and the generator comparison (Sec. V B) are given both with and without it; the note gives the per-cell uncertainties both ways too, with the block's construction and audit (Ref. [14], Sec. 10.4).

Every χ^2 of this result inverts the full covariance of the cells compared, 16×16 or 8×8, by Cholesky decomposition and without truncation, so its number of degrees of freedom is the number of cells (one fewer for the shape-only χ^2). Of the blocks, the statistical block is the unbiased sample covariance of 50 bootstrap replicas, and the flux block a covariance of 100 weighted universes (effective number 19), normalized by the sum of the weights. The others are a 2p2h band, two-point shifts, chosen models, the normalization, the run-period block, built from three exposure-weighted periods, and the training-seed block, which is zero; each comparison with a prediction adds V_g, the diagonal covariance from the finite size of the prediction's own simulated sample. No finite-ensemble correction is applied. Appendix A lists the systematic groups and gives the per-group uncertainties and the correlation matrix of the reported covariance; the note gives each block's ensemble and normalization (Ref. [14], Sec. 10.3).

B. Two-dimensional benchmark

The two-dimensional result is an end-to-end test of the unfolding against a measurement from the same data. Its integrated cross section is 3.069×10^{-38} cm^2/nucleon, compared with 3.039×10^{-38} cm^2/nucleon in the published analysis, a ratio of 1.010. The standardized bin differences formed with the published uncertainties have an RMS of 0.535. Against the full published covariance, the descriptive covariance-weighted distance is $\chi^2/\text{ndf} = 2.27$. It is computed with the pseudo-inverse over the 205 reported bins, which keeps all 205 eigenmodes of the published covariance (condition number 1.5×10^{12}), and ndf is the number of reported bins. The value is descriptive, and no p-value is quoted for it. It indicates a correlated shape difference, not a normalization difference, spread over many eigenmodes of the published covariance. The source of the shape difference has not been identified. Figure 1 shows the principal one-dimensional projections.

We quote the LightGBM result as the central value because its covariance uses the same estimator. The comparison is a consistency benchmark, not a blind validation: LightGBM, rather than the scikit-learn exact-split estimator at the same settings, was chosen after both had been compared with the published result, a choice that lowered χ^2/ndf from 3.66 to 2.65 (both with purity-based subtraction), and the distance depends on the estimator (Ref. [14], Sec. 6.4). The purity-subtracted exact-split result differs from the central value by a median of 1.03 % per bin (maximum 17.6 %; Ref. [14], Sec. 3.2).

C. Two-dimensional covariance

The two-dimensional covariance is the sum of four blocks, each from LightGBM unfoldings with the central value's estimator: systematic universes (187 universes,

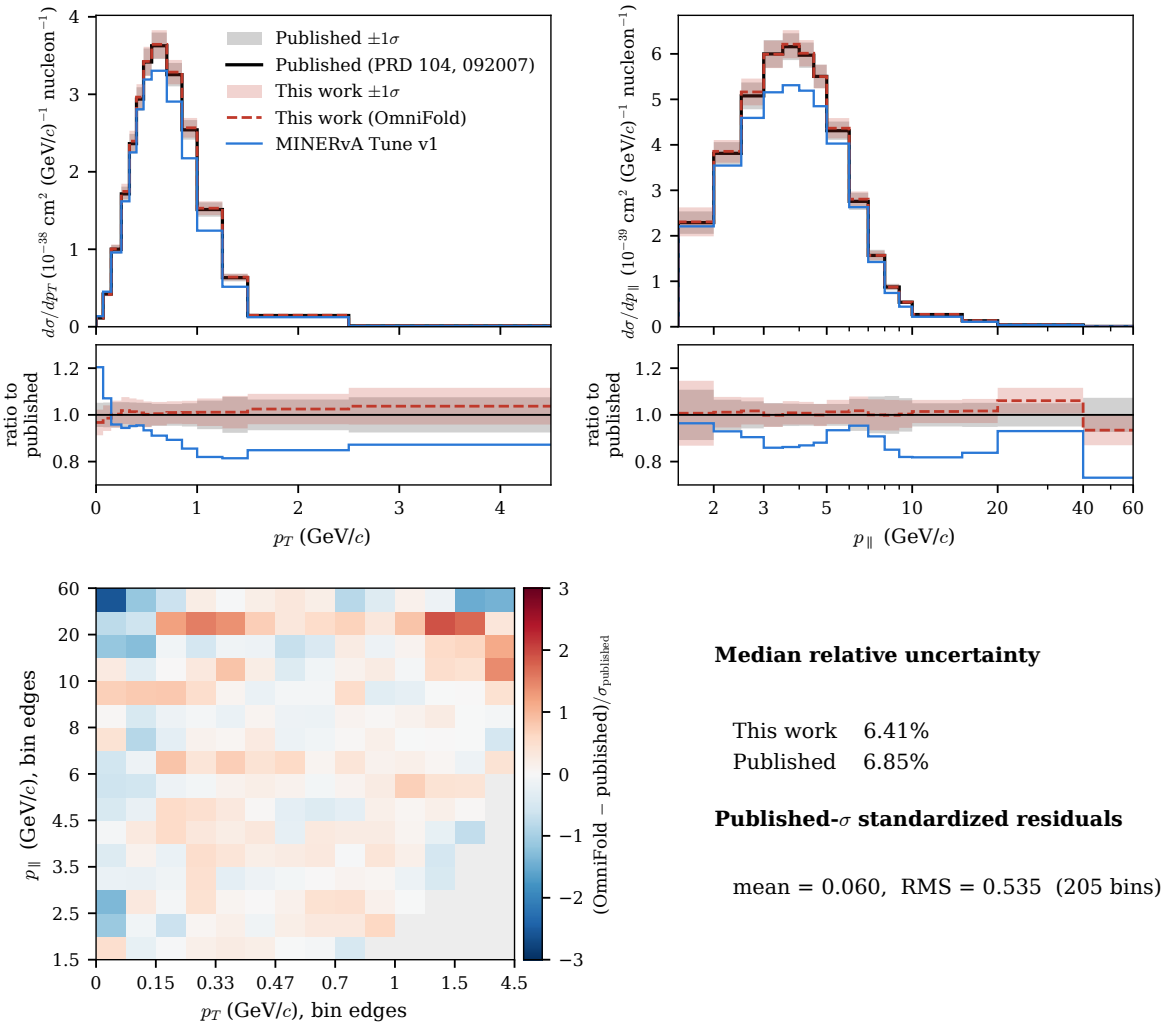


FIG. 1. Comparison with the published result [13]. Top: p_T and $p_\parallel$ projections of the published result and of this analysis, each with its total $\pm1\sigma$ band, and of the MINERvA Tune v1, with ratios to the published result below. The band of this analysis is the four-block covariance of Sec. IV C, without C_{model}. Here the two bands are correlated, so their overlap is not a compatibility test. Bottom left: (OmniFold − published)/$\sigma_{\text{published}}$ per reported bin, with cells drawn at equal size (grey: not reported). Bottom right: median relative uncertainties and residual summary.

including the normalization), statistical (300 bootstrap replicas), run period (three period groups), and training seed (five seed settings, so that its own size is poorly determined). The resulting relative uncertainty has a median of 6.41 % per bin and is 4.76 % on the integrated cross section, compared with 6.85 % and 4.61 % for the publication. The published flux uncertainty (median 4.01 %) is larger than ours (3.71 %), for a reason the released inputs do not let us determine. The covariance omits the approximation in the systematic-universe unfoldings for events that fail reconstruction, the learner dependence of Sec. IV B, and the dependence on the regularization, while the dependence on the prior's generator shape is reported beside it (below). The block sizes, the assumptions of the block sum, and the tests of the approximation are given in Ref. [14], Secs. 6.3, 4.5, and 4.6.

The statistical block, with a median statistical uncertainty of 0.676 %, is a bootstrap of 300 replicas drawn at the central value's training seed, in which the background templates are also resampled.

In the two-dimensional result, data from different run periods are not mutually consistent within the other uncertainties: one period group's unfolded $p_\parallel$ distribution, relative to its own simulation, differs from that of the earliest periods by a step of about 7 % near 8 GeV/c, of unknown cause (Sec. IV A). Its run-period block, built from these differences in the same way, includes this inconsistency, amounts to 2.0 % on the integrated cross section and passes its own consistency check by construction.

We report the dependence on the prior's generator shape as a separate covariance beside the four-block covariance, $C_{\text{model}} = \sum_k s_k s_k^{\mathsf{T}}$. Here s_k is the per-bin relative change of the result when the prior truth is reweighted by the ratio of the $(p_T, p_\parallel)$ shape of generator k (GENIE

with meson-exchange currents, NuWro, or GiBUU) to that of GENIE, with its normalization kept. Its median is 3.71 % per bin. It is a measure of generator-shape model dependence, not a calibrated bound on the unfolding bias. Added to the four-block covariance, it gives a total with a median of 7.90 % per bin. The comparison with the published uncertainty uses the covariance without C_{model}, because neither has a component for the residual dependence on the prior's generator shape. For the published analysis, this is because it chose its ten iterations in a fake-data study and assigns no uncertainty to the remaining dependence [13]. The iteration counts differ (ten there, five here), so we make no statement about the size of the published analysis's residual.

D. Normalization and closure

Summed over each of the four $(p_T, p_\parallel)$ quadrants, the sixteen-cell result agrees with the two-dimensional result: the sums differ by -1.5 to $+0.7$ per cent, and with the covariance between the two results evaluated from the systematic variations they share, the four differences give $\chi^2 \leq 1.25$ for four degrees of freedom ($p \geq 0.87$), with no quadrant beyond one standard deviation. The statistical correlation, and that of four systematic terms that could not be evaluated directly, are bounded within a one-parameter family of valid covariances, over which the agreement holds. Of the joint-test estimator's other results, only the three-observable marginal has been compared with the two-dimensional result, agreeing with it to 0.141 % in area-weighted total over all 205 reported cells (Ref. [14], Sec. 10.2). The normalization checks of the earlier purity-subtracted five-observable central value are described in Ref. [14], Sec. 8.1, and the closure checks of its estimator in three and four observables in Ref. [14], Sec. 5.1.

Closure tests with backgrounds included, comparing purity-based with unbinned negative-weight subtraction, are given in Ref. [14], Sec. 9.2 and Ref. [14], App. C.6.

E. Generator-sized truth changes

Figure 2 shows that the unfolding recovers only part of a change of the E_{avail} shape as large as the difference between generators. When the true shape is reweighted by the GiBUU-to-GENIE ratio, the (E_{avail}, W) cells unfolded with a development variant, used while the joint-test estimator was developed, with a 100-tree, 8-leaf refinement, not the reported estimator, deviate from it by several percent in the largest cells and by up to 74 % at high W, in the cell $0.8 \leq E_{\text{avail}} < 1.5\,\text{GeV}$, $W \geq 3\,\text{GeV}$. These deviations occur with or without background, so background subtraction alone does not explain the residual. In a noise-free comparison in simulation, an exact binned iterative unfolding of the same development truth changes leaves about the same median residual at five iterations; longer exact iteration removes much of the residual of the GiBUU-to-GENIE change (from 8.3 % at five iterations to 3.3 % at 200) but not of a NuWro $(p_T, p_\parallel, E_{\text{avail}})$ shape or a q_3 stress test. The residual is therefore consistent with finite-iteration regularization for the GiBUU-like change, but not for every change tested, and no single cause is established (Ref. [14], Sec. 9.3). For other fixed generator shapes tested with the joint-test estimator, it is 16 % to 31 % in the most affected (E_{avail}, W) cell and lies within the quoted total uncertainty in about half of those cells. Those uncertainties come from an earlier purity-subtracted five-observable covariance that fails its specified acceptance criteria (Ref. [14], Sec. 9.6). For the sixteen-cell result, Sec. IV G reports the fake-data tests of the estimator's bias. The deformation studies, the iteration and capacity tests, the exact binned comparison with its scope, and the joint-test-estimator shapes with the seed sets of their pseudo-experiments are given in Ref. [14], Secs. 9.2 and 9.3.

Statistical intervals of an earlier five-observable estimator with purity-based subtraction, corrected for its closure bias at nominal truth, miscover at the tested truths, most strongly when the true signal rate is lowered by about 10 %, with coverage as low as 0.06 for 68% intervals (Ref. [14], Sec. 9.5).

F. Sensitivity to input rounding

The joint-test estimator is unstable under floating-point rounding of its inputs: over 20 rounding-level perturbations of the data, the reported quantities scatter by a median of 0.23 % per joint cell, against a median data-bootstrap statistical uncertainty of 0.37 %, and each bootstrap replica carries this scatter at the same size. It limits how exactly the central values can be reproduced, and a single perturbation moves the most affected quantity further. Rounding-level changes of its inputs move its results by up to 2.6 statistical standard deviations, and those of the lower-capacity development variant by up to 2.5, so the estimator fails a data-stability requirement of at most 0.3 (median) and 1.0 (maximum) statistical standard deviations.

G. Fake-data tests of the sixteen-cell result

To compare the estimator's bias with the reported covariance when the truth differs from the simulation, we unfold pseudo-data at 12 truths and compare the mean unfolded result with each truth on the 16 cells through χ^2 with 16 degrees of freedom. The truths are not held out: they are the simulation's prior; five hybrid truths, each equal to a generator prediction in every cell of the fine grid and to the MINERvA Tune v1 simulation's event composition within each cell; NuWro shape changes in E_{avail} and in $(p_T, p_\parallel, E_{\text{avail}})$; a GENIE + MEC shape change in (E_{avail}, W); a GiBUU-to-GENIE E_{avail} ratio; a

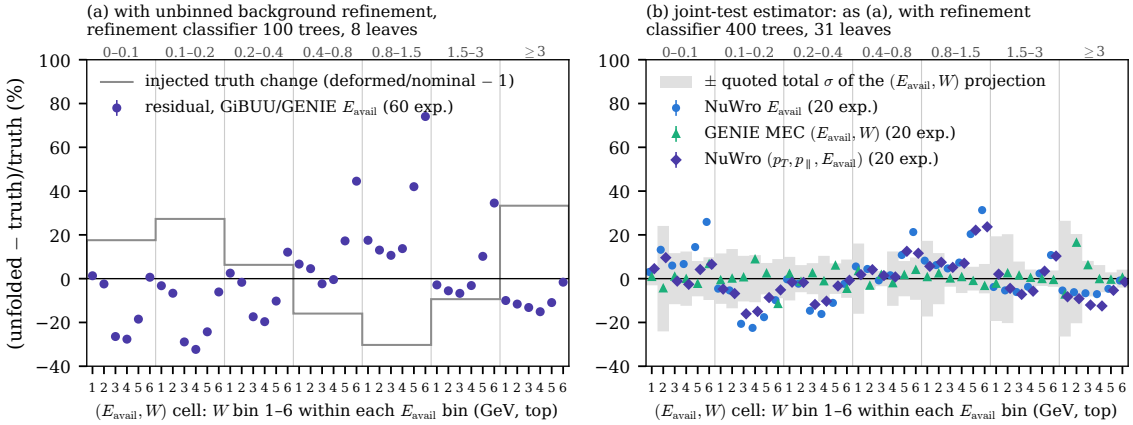


FIG. 2. Fractional recovery residual, (unfolded − truth)/truth, per (E_{avail}, W) cell, where the truth is each pseudo-experiment's deformed truth; all pseudo-experiments include backgrounds. Neither panel shows the sixteen-cell result or its reported covariance. (a) A development variant of the joint-test estimator, not the reported estimator, whose refinement has 100 trees and 8 leaves, with the truth reweighted by the GiBUU/GENIE E_{avail} shape ratio (60 pseudo-experiments), where the grey step is the injected change of the cell truth. At nominal truth this variant's largest residual is 0.39 % (Ref. [14], Sec. 9.2). (b) The joint-test estimator, for three shape changes not used while it was chosen (NuWro in E_{avail}; GENIE + MEC in (E_{avail}, W); NuWro in $(p_T, p_\parallel, E_{\text{avail}})$), 20 pseudo-experiments each, where the grey band is ± the total uncertainty of the (E_{avail}, W) projection of the covariance of an earlier purity-subtracted five-observable central value, not the reported covariance (Ref. [14], Sec. 9.6). At nominal truth (40 pseudo-experiments) its largest residual is 0.09 %. W bin 6 is $W \geq 3$ GeV. Error bars are the standard error of the mean.

noise-free closure at the NuWro shape; and a synthetic tilt of the q_3 distribution at fixed (E_{avail}, W). For a truth from a generator that contributes to C_{model}, the covariance omits that generator's component (the Cov. column of Table I). Two of the truths, the simulation prior and the Tune v1 hybrid, are the simulation's own truth, so they are closure tests (marked * in Table I). A truth is flagged when $p < 0.01$ or when a cell has $|z| > 2$; Table I lists all 12 truths. Table I gives the results under the reported covariance and under C_{total} alone, and 6 further truths, given in the note, are descriptive and not judged against these criteria.

Under the reported covariance, without the GiBUU component for the two GiBUU truths, the flagged truths ($p < 0.01$, or a cell with $|z| > 2$) are the GiBUU hybrid truth ($p < 10^{-27}$; cells R02, R06) and the synthetic q_3 tilt, by p alone ($p < 10^{-5}$). Without $C_{\text{PL,5D}}$, the GiBUU E_{avail} shape also meets the flagging criterion, in cell R06 ($z = 2.01$, against 1.76 with the block). Thus the fake-data bias exceeds the stated thresholds at the GiBUU hybrid truth and the q_3 tilt under the reported covariance, and also at the GiBUU E_{avail} shape without $C_{\text{PL,5D}}$. The bias is the same in both; only the covariance differs. At the other 10 truths the bias passes both criteria under the reported covariance, with no cell beyond 2σ and $p \geq 0.01$.

Two GENIE v3 tunes, G18_10a_02_11b and AR23_20i_00_000, were generated for this test and held out of the analysis. Each reweights the simulated truth using its shape ratio to GENIE CV, constructed as for the C_{model} components. The simulated detector response is not varied in these tests. Separate noise-free closures at the nominal truth, with pseudo-data from a simulation whose reconstructed recoil response is shifted and the unfolding on the nominal response, move the sixteen cells by at most 0.40σ when each deposit category of C_{resp} is shifted by its $\pm 1\sigma$, and by a median of 0.50σ (at most 0.97σ) when the whole recoil energy is scaled by ±4 %, with σ the per-cell uncertainty of $C_{\text{total}} + C_{\text{resp}}$. These closures are descriptive: no pass threshold is set, and they do not establish the bias under the true detector response (Ref. [14], App. D). For each tune, 20 Poisson pseudo-experiments and a noise-free closure are compared with the truth using the full reported covariance. Both tunes satisfy the bias criteria: the pseudo-experiment means give $\chi^2 = 9$ ($p = 0.90$) and 14 ($p = 0.59$), respectively, with $|z| \leq 0.62$ in every cell of the means and closures. In cell R11 at G18_10a_02_11b, the bias lies below the reported uncertainty only with C_{model} included. The tunes share a quasielastic cross-section model, and G18_10a_02_11b and the GENIE + MEC component of C_{model} also use Valencia-family MEC models. These tests provide partial evidence and are not independent. Omitting that C_{model} component gives $\chi^2 = 12$ ($p = 0.77$) and 19 ($p = 0.28$), respectively, as a descriptive comparison. Coverage is discussed in Sec. VI A.

TABLE I. Fake-data tests on the sixteen cells: χ^2 (16 degrees of freedom), p and the cells with $|z| > 2$, against the reported $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ and against C_{total}, each without the truth's own generator component where the second column says so (−). *Closure test: the truth is the simulation's own.

		$C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$			C_{total}		
Truth	Cov.	χ^2	p	cells	χ^2	p	cells
simulation prior*	full	0	1.00	—	0	1.00	—
Tune v1, hybrid*	full	0	1.00	—	0	1.00	—
GENIE CV, hybrid	full	16	0.48	—	17	0.36	—
GENIE + MEC, hybrid	−GENIE + MEC	14	0.62	—	16	0.44	—
GENIE + MEC, (E_{avail}, W) shape	−GENIE + MEC	0	1.00	—	1	1.00	—
NuWro, hybrid	−NuWro	24	0.093	—	30	0.018	—
NuWro, E_{avail} shape	−NuWro	5	1.00	—	6	0.99	—
NuWro, $(p_T, p_\parallel, E_{\text{avail}})$ shape	−NuWro	27	0.037	—	36	0.0027	—
NuWro, closure	−NuWro	11	0.83	—	13	0.67	—
GiBUU, E_{avail} shape	−GiBUU	25	0.078	—	36	0.0031	R06
GiBUU, hybrid	−GiBUU	171	$< 10^{-27}$	R02, R06	247	$< 10^{-30}$	R02, R06, R07
synthetic q_3 tilt	full	52	$< 10^{-5}$	—	60	$< 10^{-6}$	—

V. THE SIXTEEN-CELL CROSS SECTION

A. Result

Table II gives the sixteen-cell cross section: for each cell, its box in the five observables, the cell-integrated cross section f, and its marginal relative uncertainty under the reported covariance $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ of Sec. IV A. The cells are correlated, so quantitative comparisons need the full covariance, whose correlation matrix is given in Table V; Table IV gives the uncertainty of each group of the covariance, and Ref. [14], Table 12, the uncertainties without $C_{\text{PL,5D}}$.

B. Comparison on the sixteen reported cells

Unfolded results can be compared with any prediction at truth level without simulating each generator through the detector. The two-dimensional result is therefore reusable, and the sixteen-cell result can be used for the comparison that follows. Table III compares the sixteen-cell result of Sec. V with five predictions through a χ^2 against the reported covariance plus each prediction's finite-sample (diagonal) covariance V_g, with a shape-only χ^2 beside it. Figure 3 shows the data over each prediction, cell by cell. GiBUU's sample stops at $E_\nu = 20$ GeV, so GiBUU is compared on the 8 cells with $p_\parallel < 4.5$ GeV/c only. All five predictions are in the analysis definitions, with NuWro and GiBUU corrected to CH (Sec. II), and MINERvA Tune v1 is the simulation's prior. On carbon, NuWro gives $\chi^2 = 130$ (shape-only 113) and GiBUU 45 (25). For GiBUU the correction raises the total χ^2 by 2.3, from 45 to 48 for 8 degrees of freedom, and lowers the shape-only χ^2 from 25 to 22. That rise comes from pairing the CH prediction with C_{model}, whose NuWro and GiBUU shape ratios are from carbon (Sec. IV A): with those ratios also from CH, GiBUU's total instead falls below its carbon value. With the reported C_{model}, the sign of the rise also depends on the hydrogen sample: in the generators' own definitions, GiBUU's total with GENIE's hydrogen in place of its own also falls below its carbon value, but GENIE's hydrogen sample is thin, so this difference is largely its noise; the values are given in Ref. [14], Sec. 10.7. A native CH generation of NuWro agrees with the combination within a tolerance fixed in advance ($\sum z^2 = 17.9$ for 16 cells, largest $|z| = 2.2$, in the generators' own definitions). Taking C_{model}'s NuWro and GiBUU shape ratios from CH instead changes the total χ^2 by −0.9 for GENIE CV, −5.6 for GENIE + MEC, +0.2 for NuWro and −2.7 for GiBUU; the size of C_{model} is not calibrated: doubling it changes them by −2.2, −5.2, −0.8 and −2.3, and removing it by +14.6, +32.5, +12.6 and +119.0 (Ref. [14], Sec. 10.7).

With the run-period block, each total χ^2 is lower by 3.6–29.3 % of its value without it, and each shape-only χ^2 by 7.4–40.4 %. The paired columns show how the comparison depends on this block; they calibrate neither set of values and establish no exclusion of a generator.

Weakening only the flux correlations by 10 % lowers the χ^2 of the 16-cell predictions by 23–38 %, and the same change to the detector or lateral correlations lowers it by at most 18 %. With the correlations of the interaction, flux, detector and lateral groups removed, every 16-cell prediction keeps $\chi^2 \geq 41$ for 16 degrees of freedom. The flux correlations are estimated from 100 hadron-production (PPFX) flux universes [16], weighted by the neutrino–electron and inverse-muon-decay constraint. Bootstrapping these universes, with the flux variances kept, gives 16th–84th percentile bands for each 16-cell χ^2, with edges within 3.8 % of the reported value and a median at most 0.4 % above it. The bands describe the scatter of the χ^2 from estimating the correlations with these universes (Ref. [14], Sec. 10.6).

VI. DISCUSSION AND CONCLUSIONS

The principal result is the sixteen-cell cross section of Sec. V, with its covariance and generator comparison.

Two validation findings point to the unfolded result retaining part of its prior. The first is the residual under a

TABLE II. The sixteen-cell cross section: each cell's box (p_T, $p_\parallel$ and q_3 in GeV/c; E_{avail} and W in GeV), the cell-integrated cross section f (an integral over the box, not a differential density) in units of $10^{-40}\,\text{cm}^2$/nucleon, and the marginal relative uncertainty σ/f (%), the square root of the diagonal of $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ over f. The largest edge on each axis is the outer edge of the analysis grid, which contains every signal event (Ref. [14], Sec. 3.5). The last row gives the median over the cells. Quantitative comparisons need the full covariance (Sec. IV A). †Reconstruction stability 0.16–0.25 (Sec. VI A, item 2).

cell	p_T	$p_\parallel$	E_{avail}	q_3	W	f	marginal σ/f
R00	0–0.7	1.5–4.5	0–0.8	0–2	0–100	32.6	7.11
R01	0–0.7	1.5–4.5	0–100	2–100	1.8–100	36.7	7.59
R02†	0–0.7	1.5–4.5	0.8–100	0–2	0–1.8	6.31	14.05
R03	0–0.7	1.5–4.5	0.8–100	0–2	1.8–100	4.30	12.73
R04	0–0.7	4.5–60	0–0.8	0–2	0–100	49.0	4.80
R05	0–0.7	4.5–60	0–100	2–100	1.8–100	9.98	8.12
R06†	0–0.7	4.5–60	0.8–100	0–2	0–1.8	5.57	6.56
R07	0–0.7	4.5–60	0.8–100	0–2	1.8–100	3.21	8.45
R08	0.7–4.5	1.5–4.5	0–0.8	0–100	0–1.8	13.9	6.26
R09	0.7–4.5	1.5–4.5	0–100	0–100	1.8–100	39.6	6.85
R10†	0.7–4.5	1.5–4.5	0.8–100	0–2	0–1.8	7.83	8.69
R11	0.7–4.5	1.5–4.5	0.8–100	2–100	0–1.8	5.61	13.89
R12	0.7–4.5	4.5–60	0–0.8	0–2	0–100	22.0	6.98
R13	0.7–4.5	4.5–60	0–100	2–100	0–1.8	9.04	11.38
R14	0.7–4.5	4.5–60	0–100	2–100	1.8–100	49.6	5.68
R15†	0.7–4.5	4.5–60	0.8–100	0–2	0–100	11.3	8.29
median							7.86

TABLE III. χ^2 for each prediction in the indicated comparison region, against the reported covariance $C_{\text{total}} + C_{\text{resp}} + C_{\text{PL,5D}}$ and against the covariance without $C_{\text{PL,5D}}$, each plus V_g, where V_g is the prediction's finite-sample (diagonal) covariance. Shape-only χ^2 frees the normalization. All five predictions are in the analysis definitions, with NuWro and GiBUU corrected to CH (Sec. II). The ndf of each block is given in its heading.

	reported covariance		without $C_{\text{PL,5D}}$	
Prediction	χ^2	shape-only χ^2	χ^2	shape-only χ^2
All 16 cells: ndf 16, shape-only 15				
GENIE 2.12.10 CV	107	102	111	110
GENIE 2.12.10 + Valencia MEC	229	229	278	264
NuWro 21.09 (CH)[a]	128	108	181	181
MINERvA Tune v1	262	262	330	311
8-cell subset, $p_\parallel < 4.5$ GeV/c: ndf 8, shape-only 7				
GiBUU 2019 (CH)[b]	48	22	61	25

[a] In NuWro's free-proton sample, the supplement generated at neutrino energies of 50–100 GeV is flagged: the events lost to generator aborts there were not shown to be negligible (Ref. [14], Sec. 11.9).

[b] Corrected to CH with GiBUU's own free-proton sample, which enters with weight 1/13 per nucleon. That sample failed its sampling-closure check, and its deep-inelastic events, 46 % of it, do not conserve energy (Ref. [14], Sec. 11.9).

generator-sized change of the truth in Sec. IV E, which for the GiBUU-like change is consistent with finite-iteration regularization. The second is the shift of the data result when the simulation's prior is reweighted to generator-sized changes of the truth, which tracks the same bias (Ref. [14], Sec. 9.4). The note's calibrated tests show the same bias toward the prior: it displaces their null distributions for the external predictions upward, and the calibration, which simulates the whole analysis chain, reproduces that displacement (Ref. [14], Sec. 11.5).

The studies of Sec. IV and of the companion note show what must be checked before unfolded intervals are interpreted: closure with the backgrounds included; truth changes of generator size, not only nominal closure; coverage under changes of the truth, not only at nominal truth; robustness to training seeds and rounding-level input changes; and run-period consistency.

A. Limitations

Apart from the validation of the uncertainties and the low stability of four of the sixteen cells (below), the limitations of these results are stated with the results they qualify, in Secs. III–V, and those of the calibrated tests in the companion note. The principal ones are these.

1. No reported interval has validated coverage. The fake-data tests of Sec. IV G assess bias, not coverage: only event counts fluctuate, with systematic parameters fixed, so interval widths cannot be tested. A coverage test at two of their truths, declared in advance for the covariance without the run-period block, with pseudo-experiments that also draw the flux, interaction, and detector uncertainties, was stopped by one of its preliminary checks before any coverage pseudo-experiment ran: the uncertainties reproduced through its pseudo-experiments disagreed with the covariance blocks they draw, for reasons not established (Ref. [14], Sec. 10.5). Neither the coverage of C_{model} at truths held out of its construction nor the size of either run-period block is independently validated (Ref. [14], Secs. 10.3, 10.4 and 4.5). The two-dimensional covariance has not been independently validated and gives no cal-

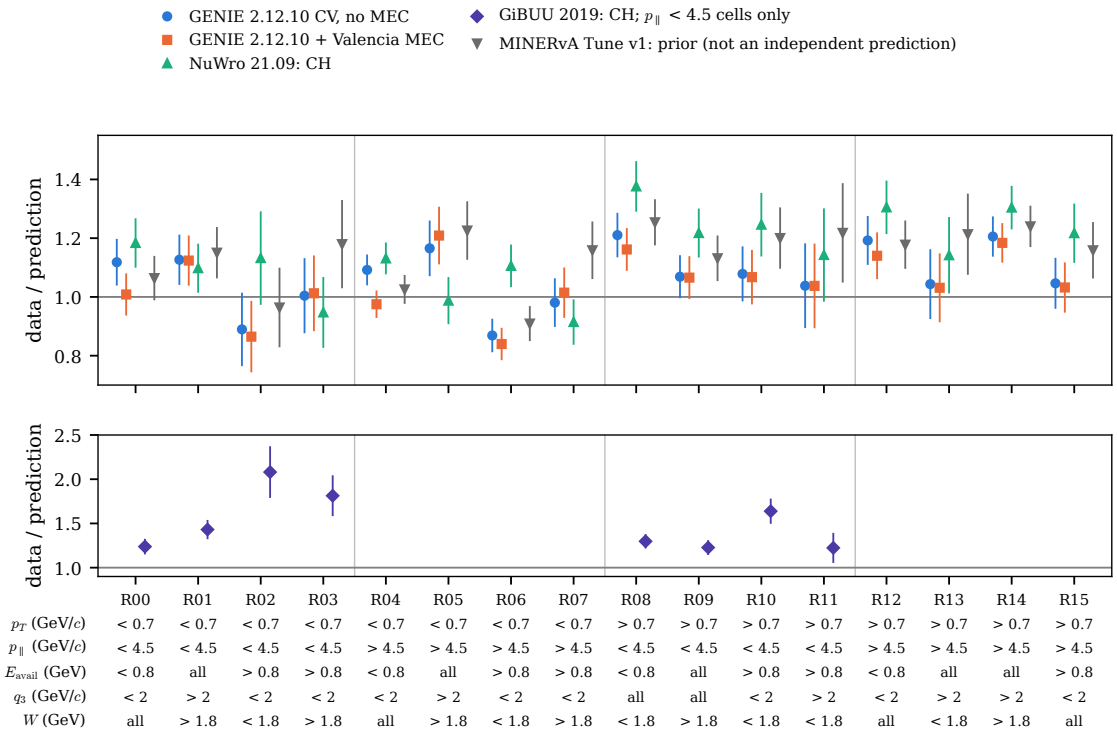


FIG. 3. Data over prediction, f/g, on the sixteen cells for the five predictions of Table III, all in the analysis definitions, with NuWro and GiBUU corrected to CH. Each error bar is $(f/g)\,\sigma/f$, with σ the square root of the diagonal of $C_{\text{total}}+C_{\text{resp}}+C_{\text{PL,5D}}$, and the predictions' finite-sample uncertainties are not drawn. GiBUU (lower panel) is shown on its 8 cells with $p_\parallel < 4.5$ GeV/c only. The key below the axes gives each cell's box. The χ^2 of Table III uses the full covariance, whose correlations the error bars do not show.

ibrated significance for generator comparisons; an earlier statistical band of that result undercovered in its tails in a coverage test at fixed truth (Ref. [14], Sec. 5.2), and the coverage of the current statistical block and of the total covariance has not been tested. The coverage tests cited are pointwise, for each reported quantity separately, not the stronger simultaneous coverage of all of them. We make no coverage claim on independent simulated samples or simultaneously over all reported bins and projections.

2. Four of the sixteen cells, R02, R06, R10 and R15, have a reconstruction stability of 0.16–0.25: of the simulated signal events generated in such a cell and reconstructed, only that fraction is reconstructed in the same cell.

3. The note's calibrated tests reject hybrid nulls, conditionally on the simulation's hadronic response. However, a rejection extends to the generator prediction itself only if the residual below the fine grid, where each null takes the simulation's event composition, is bounded, and no such bound has been shown. Their calibration is valid to the extent that the simulated chain reproduces the analysis of the data (Ref. [14], Sec. 11.4). They are not physical exclusions of the generators, and they show no evidence of additional discrimination beyond matched coarse projections.

Appendix A: Uncertainty tables

The reported covariance of the sixteen-cell result (Sec. IV A) is a sum of blocks, grouped as follows.

- *Statistical*: the bootstrap of the data, 50 replicas.
- *Flux*: 100 hadron-production (PPFX) flux universes, weighted by the neutrino–electron and inverse-muon-decay constraint (effective number 19).
- *Interaction*: the 33 interaction-model bands of the MINERvA systematic set, which reweight simulated events.
- *Detector*: the 4 detector-model bands, which reweight simulated events: the MINOS muon-matching efficiency and the GEANT4 neutron, pion and proton reweights.
- *Lateral*: the 5 bands that shift the reconstructed kinematics of each simulated event rather than reweighting it: the beam angle in two directions, the muon momentum resolution, and the muon energy scales in MINERvA and in MINOS.
- *Normalization*: 1.40 % in every cell.
- C_{model}, C_{resp} and $C_{\mathrm{PL,5D}}$, as defined in Sec. IV A.

The statistical, flux, interaction, detector, lateral and normalization groups, with a training-seed block that is zero in every cell, form C_{e}, and $C_{\mathrm{total}} = C_{\mathrm{e}} + C_{\mathrm{model}}$. Table IV gives each group's relative uncertainty in each cell, and Table V the correlation matrix of the reported covariance. With the cross sections and the marginal uncertainties of Table II, the correlation matrix gives the reported covariance to the printed precision; the covariance is also provided in the Supplemental Material.

ACKNOWLEDGMENTS

The authors thank the MINERvA Collaboration for making their data, simulated data, and analysis tools available to the community [12, 15], and the authors of the MINERvA Analysis Toolkit and the MINERvA-101 cross-section framework on which this analysis builds. We thank Callum Wilkinson for connecting us with this dataset and for many helpful discussions, and Kevin McFarland for suggesting a detector-response-mismatch closure targeting correlations among the reconstructed hadronic observables. This research used resources of the National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility, under project m3246. [HOLD: funding and grant acknowledgments] AI language-model tools (Anthropic Claude and OpenAI GPT models, used through the Claude Code and OpenAI Codex agents) gave substantial assistance with the analysis code, statistical procedures, computing campaigns, independent recomputations and reviews, figures, and manuscript drafts. The authors directed the work and verified the results; the analysis note [14] gives the full statement, including the tool versions. The figures were drawn by plotting code from the analysis outputs; no figure was generated or edited by a generative-AI image tool. AI tools are not authors, and the authors take responsibility for the content.

DATA AVAILABILITY

The input data and simulation are the MINERvA open-data AnaTuples, released under CC0 [12]. The sixteen-cell cross section, its covariance matrix and the cell definitions are provided as machine-readable files in the Supplemental Material. This analysis used an earlier production of that release; its 2,374 input files are identified by SHA256 checksum in the records cited in Ref. [14], App. G. The analysis note's figures of the joint-test null distributions and the (E_{avail}, W) map are regenerated by a candidate release package of about 16 MB, not yet deposited, that requires only numpy, scipy, and matplotlib. The paper's quoted numbers and the event-level processing require the archived analysis products, the open-data AnaTuples, and the generator samples produced for this analysis; Ref. [14], App. G states what each part of the release does and does not reproduce. [HOLD: deposit identifier and access route, after the separately authorized release decision.]

[1] G. D'Agostini, A Multidimensional unfolding method based on Bayes' theorem, Nucl. Instrum. Meth. A **362**, 487 (1995).

[2] A. Andreassen, P. T. Komiske, E. M. Metodiev, B. Nachman, and J. Thaler, OmniFold: A Method to Simultaneously Unfold All Observables, Phys. Rev. Lett. **124**, 182001 (2020), arXiv:1911.09107 [hep-ph].

[3] A. Andreassen, P. T. Komiske, E. M. Metodiev, B. Nachman, A. Suresh, and J. Thaler, Scaffolding Simulations with Deep Learning for High-dimensional Deconvolution, ICLR simDL workshop (2021), arXiv:2105.04448 [stat.ML].

[4] H1 Collaboration, Measurement of Lepton-Jet Correlation in Deep-Inelastic Scattering with the H1 Detector Using Machine Learning for Unfolding, Phys. Rev. Lett. **128**, 132002 (2022), arXiv:2108.12376 [hep-ex].

[5] G. Aad *et al.* (ATLAS), A simultaneous unbinned differential cross section measurement of twenty-four Z+jets

TABLE IV. Per-cell relative uncertainty (%) of each group of the reported covariance: the square root of the group's diagonal over f. C_{e} is the sum of the first six groups; the last column is the reported $C_{\mathrm{total}} + C_{\mathrm{resp}} + C_{\mathrm{PL,5D}}$, as in Table II. The last row gives the median over the cells.

cell	stat.	flux	interaction	detector	lateral	norm.	C_{e}	C_{model}	C_{resp}	$C_{\mathrm{PL,5D}}$	reported
R00	0.20	3.93	0.98	1.56	5.01	1.40	6.77	0.43	2.07	0.40	7.11
R01	0.23	3.58	1.18	2.14	4.85	1.40	6.66	0.67	2.35	2.71	7.59
R02	0.33	3.75	3.56	1.96	5.08	1.40	7.65	11.36	2.95	1.14	14.05
R03	0.33	3.52	5.39	1.66	4.70	1.40	8.27	9.38	1.85	1.50	12.73
R04	0.14	3.47	0.58	1.31	0.57	1.40	4.05	0.88	0.70	2.31	4.80
R05	0.37	4.14	2.13	1.68	1.85	1.40	5.48	3.46	3.71	3.16	8.12
R06	0.33	3.48	2.25	1.21	0.73	1.40	4.61	3.19	1.81	2.88	6.56
R07	0.30	3.90	5.08	1.76	1.75	1.40	7.02	3.45	1.55	2.79	8.45
R08	0.22	3.74	1.52	2.17	0.77	1.40	4.86	1.63	3.58	0.28	6.26
R09	0.18	3.52	1.85	2.18	1.08	1.40	4.87	3.24	2.26	2.77	6.85
R10	0.25	3.42	5.53	2.42	0.43	1.40	7.10	4.03	2.69	1.29	8.69
R11	0.31	3.35	5.16	2.56	2.91	1.40	7.42	11.15	2.85	2.36	13.89
R12	0.18	3.48	1.95	1.68	4.25	1.40	6.23	0.51	2.20	2.18	6.98
R13	0.28	3.85	5.93	2.08	3.93	1.40	8.47	6.81	2.08	2.65	11.38
R14	0.16	3.73	1.22	2.14	1.34	1.40	4.87	0.82	1.82	2.13	5.68
R15	0.22	3.48	3.19	1.82	3.16	1.40	6.13	4.78	1.64	2.37	8.29
median	0.24	3.55	2.19	1.89	2.38	1.40	6.45	3.34	2.14	2.34	7.86

TABLE V. Correlation matrix of the reported covariance $C_{\mathrm{total}} + C_{\mathrm{resp}} + C_{\mathrm{PL,5D}}$ on the sixteen cells (lower triangle; the diagonal is one).

	R00	R01	R02	R03	R04	R05	R06	R07	R08	R09	R10	R11	R12	R13	R14
R01	0.69														
R02	0.52	0.30													
R03	0.53	0.32	0.81												
R04	0.49	0.53	0.15	0.21											
R05	0.34	0.70	−0.03	−0.14	0.57										
R06	0.36	0.51	0.64	0.57	0.59	0.43									
R07	0.41	0.40	0.40	0.75	0.55	0.19	0.64								
R08	0.66	0.30	0.24	0.31	0.73	0.13	0.31	0.40							
R09	0.19	0.57	−0.24	−0.07	0.77	0.65	0.42	0.48	0.40						
R10	0.37	0.13	0.73	0.42	0.41	0.16	0.56	0.21	0.36	−0.06					
R11	0.06	0.07	−0.40	−0.43	0.39	0.46	−0.04	−0.15	0.16	0.26	0.14				
R12	0.02	0.03	−0.01	0.00	0.73	0.22	0.35	0.26	0.60	0.51	0.29	0.34			
R13	0.03	0.05	−0.26	−0.38	0.51	0.50	0.13	−0.09	0.21	0.31	0.34	0.92	0.46		
R14	0.24	0.52	0.18	0.21	0.74	0.65	0.65	0.48	0.38	0.72	0.26	0.11	0.64	0.25	
R15	0.08	0.05	0.61	0.43	0.52	0.11	0.71	0.41	0.32	0.20	0.83	−0.04	0.54	0.23	0.52

kinematic observables with the ATLAS detector, Phys. Rev. Lett. **133**, 261803 (2024), arXiv:2405.20041 [hep-ex].

[6] F. Canelli *et al.*, A Practical Guide to Unbinned Unfolding, Eur. Phys. J. C **86**, 106 (2026), arXiv:2507.09582 [hep-ph].

[7] R. G. Huang *et al.*, Machine learning assisted unfolding for neutrino cross-section measurements with the OmniFold technique, Phys. Rev. D **112**, 012008 (2025), arXiv:2504.06857 [hep-ex].

[8] D. Ruterbories *et al.* (MINERvA), Simultaneous measurement of proton and lepton kinematics in quasielasticlike ν_μ-hydrocarbon interactions from 2 to 20 GeV, Phys. Rev. Lett. **129**, 021803 (2022), arXiv:2203.08022 [hep-ex].

[9] D. Ruterbories *et al.* (MINERvA), Comparisons of triple-differential cross sections for quasielastic-like ν_μ-hydrocarbon interactions using $\langle E_\nu \rangle \sim 3$ GeV versus ~ 6 GeV beams in MINERvA (2026), arXiv:2606.00745 [hep-ex].

[10] P. Abratenko *et al.* (MicroBooNE), Measurement of three-dimensional inclusive muon-neutrino charged-current cross sections on argon with the MicroBooNE detector, Phys. Lett. B **870**, 139939 (2025), arXiv:2307.06413 [hep-ex].

[11] M. A. Acero *et al.* (NOvA), First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector (2026), arXiv:2603.06718 [hep-ex].

[12] MINERvA Collaboration, MINERvA Open Data Product, Fermilab, `https://minerva.fnal.gov/opendata/`, doi:10.15484/3022562, released under CC0.

[13] D. Ruterbories *et al.* (MINERvA), Measurement of inclusive charged-current ν_μ cross sections as a function of muon kinematics at a mean neutrino energy of 6 GeV on hydrocarbon, Phys. Rev. D **104**, 092007 (2021), arXiv:2106.16210 [hep-ex].

[14] J. Bailey and B. Nachman, Unbinned OmniFold analysis of MINERvA charged-current inclusive data: analysis note, MINERvA-OmniFold repository, `https://github.com/josephbaileyy/MINERvA-OmniFold/tree/main/docs/analysis-note` (2026), review version; the archival identifier will replace this link when the note is deposited.

[15] L. Aliaga *et al.* (MINERvA), Design, Calibration, and Performance of the MINERvA Detector, Nucl. Instrum. Meth. A **743**, 130 (2014), arXiv:1305.5199 [physics.ins-det].

[16] L. Aliaga *et al.* (MINERvA), Neutrino flux predictions for the NuMI beam, Phys. Rev. D **94**, 092005 (2016), arXiv:1607.00704 [hep-ex].

[17] C. Andreopoulos *et al.*, The GENIE Neutrino Monte Carlo Generator, Nucl. Instrum. Meth. A **614**, 87 (2010), arXiv:0905.2517 [hep-ph].

[18] P. A. Rodrigues *et al.* (MINERvA), Identification of Nuclear Effects in Neutrino-Carbon Interactions at Low Three-Momentum Transfer, Phys. Rev. Lett. **116**, 071802 (2016), arXiv:1511.05944 [hep-ex].

[19] M. V. Ascencio *et al.* (MINERvA), Measurement of inclusive charged-current ν_μ scattering on hydrocarbon at $\langle E_\nu \rangle \sim 6$ GeV with low three-momentum transfer, Phys. Rev. D **106**, 032001 (2022), arXiv:2110.13372 [hep-ex].

[20] S. Henry, H. Su, *et al.* (MINERvA), Measurement of Electron Neutrino and Antineutrino Cross Sections at Low Momentum Transfer, Phys. Rev. D **109**, 092008 (2024), arXiv:2312.16631 [hep-ex].

[21] J. Nieves, I. Ruiz Simo, and M. J. Vicente Vacas, Inclusive Charged-Current Neutrino-Nucleus Reactions, Phys. Rev. C **83**, 045501 (2011), arXiv:1102.2777 [hep-ph].

[22] T. Golan, J. T. Sobczyk, and J. Żmuda, NuWro: the Wroclaw Monte Carlo Generator of Neutrino Interactions, Nucl. Phys. B Proc. Suppl. **229-232**, 499 (2012).

[23] O. Buss, T. Gaitanos, K. Gallmeister, H. van Hees, M. Kaskulov, O. Lalakulich, A. B. Larionov, T. Leitner, J. Weil, and U. Mosel, Transport-theoretical Description of Nuclear Reactions, Phys. Rept. **512**, 1 (2012), arXiv:1106.1344 [hep-ph].

[24] B. Nachman and J. Noll, Stay Positive: Neural Refinement of Sample Weights (2025), arXiv:2505.03724 [physics.data-an].

[25] F. Pedregosa *et al.*, Scikit-learn: Machine Learning in Python, J. Mach. Learn. Res. **12**, 2825 (2011), arXiv:1201.0490 [cs.LG].

[26] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A Highly Efficient Gradient Boosting Decision Tree, in *Advances in Neural Information Processing Systems 30* (2017).

method-dependence studies (§6.4).

The estimator configurations. Six estimator configurations, labeled (a)–(f), are used; Table 2 gives each one's learner, background treatment and results, and each result names the configuration it uses. All use five OmniFold iterations and gradient-boosted trees of nominal capacity (100 trees, 8 leaves) in both OmniFold steps; "deterministic" means fixed seeds and reproducible training. Results from different configurations are not combined. Four are described in this section:

- (f), the nominal LightGBM configuration above with refined negative-weight subtraction and the refinement classifier of (e), seeded at the estimator seed plus three: the 2D central value (seed 42) and its uncertainty ensembles (§6);
- (a), the same learner with purity-based background subtraction (§3.1): the five-dimensional central value (§8), the 3D and 4D injected-bump closures and Figs. 17 and 23 (§7), and the purity-subtracted 2D comparisons of this section and §6.4;
- (e), the joint-test estimator: a deterministic LightGBM variant with the unbinned negative-weight background treatment (App. C), whose refinement classifier has 400 trees and 31 leaves. The negative-weight treatment reduces the purity method's nominal-truth bias in the highest-W column at least fifteen-fold (from 3.5 % to 4.1 % to 0.11 % to 0.24 %, the level of the no-background closure), and the larger refinement classifier removes most of the background-related bias that remains (§§9.2–9.3). It gives the 3D and 4D central values (§7) and the sixteen-cell five-dimensional result (§10). It is used for the generator tests (§11), the study of generator shapes held out of its development (§9.3) and the precision study (§9.4). It was selected before the joint-test design was fixed (§11);
- (b), the exact-split cross-check above.

Configuration (c) is the deterministic variant with purity-based subtraction, and (d) is the same variant with negative weights and a nominal-capacity refinement; they serve the background-bias, generator-sized-change and coverage studies listed in the table. The five-dimensional central value of §8 and its five-axis covariance use configuration (a): they were produced before the purity bias was found and the refinement was developed. A refined five-dimensional data unfolding (configuration (e)) gives the sixteen-cell result; its reported covariance on the sixteen cells is described in §10.3. The 2D central value and its covariance were rebuilt with configuration (f) after a two-dimensional closure test with backgrounds found the purity bias (App. C.6). The change was decided on that closure test, before the rebuilt result's distance from the published result was computed, and the rebuilt result was adopted after that distance was known. The exact-split central value was the quoted 2D central value until 2026-10-11; studies labeled *exact-split* use it and have not been recomputed. The 3D and 4D central values have been regenerated with configuration (e); no 3D or 4D covariance is quoted.

Hyperparameter choice. The learner settings were fixed *a priori*; no hyperparameter search was performed on data, which avoids tuning on the result. They are the scikit-learn `GradientBoosting` defaults: 100 trees, learning rate 0.1, and depth-3 capacity. The LightGBM learner of the central value and its ensembles uses matched capacity (`num_leaves` $= 8 = 2^3$), giving every learner the same deliberately modest capacity, one of OmniFold's regularizers (§3.1). The configuration is instead validated after the fact by closure and classifier-calibration tests (§5, §7.5), insensitivity to doubling the iteration count, measured for the exact-split cross-check (below), backend agreement in method-dependence studies (§6.4), and an explicit training-seed variation band for residual ML stochasticity (§4).

an uncertainty component. Its first stage, the reconstruction-only test below, has been implemented but not yet run: no output of it has been produced and no pass threshold has been set. Closures of the same kind, from the shifted response samples of C_{resp}, have been run (below).

Reconstruction-only test. The test uses the ordinary five-dimensional closure sample. After the pseudo-data arrays are copied, and before background mixing or classifier training, it changes only the reconstructed pseudo-data record

$$E_{\text{avail}}^{\text{pseudo}} \longrightarrow (1+\epsilon) E_{\text{avail}}^{\text{pseudo}}. \tag{32}$$

The nominal reconstructed simulation, all truth coordinates, event weights, and event membership stay fixed. The pseudo-data therefore have a deliberately incompatible conditional response; OmniFold still uses the nominal response. Both signs of ϵ are required, at an amplitude fixed before the results are examined. A documented hadronic-response variation or an explicit quantitative assumption should justify that amplitude; the overall factor in the reconstructed-E_{avail} definition, Eq. (2), is not by itself a sufficient basis.

Outputs include the full five-dimensional unfolded/reference residual, one-dimensional E_{avail} and W projections, and an (E_{avail}, W) ratio map. The primary observable is the signed bias versus W, with $W < 1.1\,\text{GeV}$ reported separately from the high-W tail. The global integral is a control rather than the decision statistic. The implementation runs only as a pseudo-data closure and only on the five observables, and cannot be combined with truth-reweight injection or bootstrapping. It labels every output as a diagnostic that must not be quoted.

Closures from the shifted response samples. Noise-free, signal-only closures at the nominal truth take pseudo-data from simulation whose reconstructed recoil response is shifted, either for each deposit category of C_{resp} (§10.3) at its $\pm 1\sigma$ or as a $\pm 4\,\%$ scale of the whole recoil energy, and unfold them with configuration (e) on the unshifted response. Each closure's bias, less that of the unshifted closure, is compared on the sixteen cells with the per-cell σ of $C_{\text{total}} + C_{\text{resp}}$. For the deposit categories the median $|\Delta|/\sigma$ is 0.07–0.20 and the largest 0.40; for the whole-recoil scale the median is at most 0.50 and the largest 0.97. The closures are descriptive and carry no noise estimate: no pass threshold is set, because the closure settings were deferred. They do not establish the bias under the true detector response, validate C_{resp} or its category sizes, or describe response variants other than those tested.

Physically correlated follow-up. The reconstruction-only test is by design a computationally inexpensive adversarial stress test; it is not a detector model. A more faithful follow-up would select quasielastic events, shift the leading-nucleon energy up and down, and recompute reconstructed E_{avail}, q_3, W, and any affected selection from the altered low-level record. The current per-event outputs hold only scalar observables, not that low-level record, so the follow-up requires dedicated diagnostic output from the event processing. It is deferred unless the reconstruction-only test shows material bias or the collaboration asks for the stronger response model.

Interpretation. The test leaves the quoted central value and the current covariance construction unchanged. A small observed bias would support stability of the (E_{avail}, W) interpretation only within the tested response envelope. A large bias would require a decision on the response model or an uncertainty component, and would not be absorbed silently into a non-closure error.